

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Sixth Semester B.E. (E&C) (Credit System) Degree Examinations**

May - June 2023

20ECE86 – PYTHON PROGRAMMING

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Unit – I**Marks BT* CO* PO***

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|-------|---|----|-----|---|---|
| 1. a) | What are the key characteristics of Python programming language? How do they make it different from other programming languages? | 6 | L*2 | 1 | 1 |
| b) | How does Python work behind the scenes? Explain the Python interpreter and the bytecode it generates. | 6 | L2 | 1 | 1 |
| c) | What are docstrings in Python? How do you use them to document your code? Explain with a sample Python program using a function call related to a simple calculator. The function should receive three parameters from the primary program operator, num1, and num2, and return the result to the main program. | 8 | L3 | 1 | 1 |
| 2. a) | Explain the syntax of a for loop and how to use it in Python. How to come out of a for loop?. Write a Python program that uses a for loop to iterate over a list of numbers till a negative number is encountered and print only the prime numbers. | 10 | L3 | 2 | 1 |
| b) | Write a Python program that uses a while loop to prompt the user to enter a series of integers. For each integer, the program should perform the following actions:
i. Convert the integer to a float.
ii. If the float exceeds 100, print a message indicating that the value exceeds the threshold.
iii. If the float is less than or equal to 100, but greater than 50 multiply it by two and print the result.
iv. If the float is less than 50, skip it and move on to the next integer. | 10 | L3 | 2 | 1 |
| 3. a) | Explain the concept of short-circuit evaluation in Python. How does it impact the evaluation of complex Boolean expressions? Explain the same by taking an example of 'and' and 'or' operators to explain the flow. | 6 | L2 | 2 | 1 |
| b) | Write a Python program that prompts the user to enter two boolean values and then uses the and, or, and not operators to evaluate the following expressions:

a and b
a or b
not a

For each expression, your program should print out the input values and the result of the expression. Use conditional statements to handle cases where the user enters invalid input. | 6 | L3 | 1 | 1 |
| c) | Write a Python program that prompts the user to enter a sentence and then counts the number of occurrences of each letter in the sentence. Note: consider Upper and Lower cases as same. | 8 | L3 | 3 | 1 |

Unit – II

4. a) Write a Python program that takes a filename as input and returns the sum of all the numbers in the file. The file contains one integer per line. For example, if the file contains the numbers 1, 2, and 3, the function should return 6. 6 L3 3 1
- b) What is a dictionary in Python, and how does it differ from a list? How do you add, access, and remove keys in a dictionary? 6 L2 3 1
- c) Write a Python program that takes two inputs: a string representing a number and an integer representing the base of the number system. The program should display the integer value of the number in base 10. For example, if the input string is "1010" and the base is 2, the output should be 10. Also, explain the steps. 8 L3 3
5. a) Write a Python program that takes two lists as input and displays a list of the common elements between the two lists. For example, if the input lists are [1, 2, 3, 4] and [3, 4, 5, 6], the output should be [3, 4]. 8 L3 3 1
- b) Write a Python program that creates a dictionary of employee data, with the employee's name as the key and their age and salary as the values. The program should then prompt the user to enter an employee's name and display the corresponding age and salary information. The program should display an appropriate error message if the entered name is not found in the dictionary.

Input:

The program should first ask the user to enter the number of employees they want to enter data for.

The user should enter each employee's name, age, and salary.

Output:

After all employee data is entered, the program should prompt the user to enter an employee name.

The program should display the corresponding age and salary if the entered name is found in the dictionary.

If the entered name is not found in the dictionary, the program should display "Employee not found".

Example:

Enter the number of employees: 3

Enter employee name: John

Enter employee age: 32

Enter employee salary: 55000

Enter employee name: Jane

Enter employee age: 27

Enter employee salary: 45000

Enter employee name: Tom

Enter employee age: 41

Enter employee salary: 75000

Enter employee name to search: John

Employee Age: 32

Employee Salary: 55000

Enter employee name to search: Jane

Employee Age: 27

Employee Salary: 45000

Enter employee name to search: Bob

Employee not found

12 L3 3 1

6. a) What is recursion? Write a Python function to calculate the factorial of a given number using recursion. The function should take the input number as an argument and return the factorial. Also, write the code without recursion to show the difference.
- b) What are different methods available in Python to handle lists? Explain any four of them. Also, write a Python function called 'find_replace_string' that takes three strings as input: the input string, the string to find, and the string to replace the found string. The function should return the modified string with all occurrences of the search string replaced with the replacement string. For example, if the input string is "hello world" and we want to replace all occurrences of "o" with "a", the output should be "hella world".

10 L3 4 1

10 L3 4 1

Unit – III

7. a) Explain polymorphism with a programming example.
- b) What is an abstract class?. Give an example.
- c) Design a class called "Rectangle" that has attributes "width" and "length". The class should have methods to calculate the area and perimeter of the rectangle. The program should prompt the user to enter the width and height of the rectangle, create an instance of the Rectangle class with the entered values, and then print the area and perimeter of the rectangle.

8 L2 5 1

6 L2 4 1

Input:

The width of the rectangle

The height of the rectangle

Output:

The area of the rectangle

The perimeter of the rectangle

Note: Area of rectangle= lw and perimeter of a rectangle= $2*(l+w)$

6 L3 5 1

8. a) Write a Python program that takes a list of integers as input from the user and calculates the average of the integers. If the user enters a non-integer value, the program should print an error message and ask the user to enter an integer value. The program should use a try block to handle the exception that is raised when the user enters a non-integer value.

Input:

A list of integers.

Output:

The average of the integers.

Example:

Enter a list of integers (separated by spaces): 1 2 3 4 5 a

Error: Please enter an integer.

Enter a list of integers (separated by spaces): 1 2 3 4 5

The average of the integers is 3.0.

Also, explain the flow of the code.

8 L3 4 1

- b) How do you create a simple GUI using Python's Tkinter library?

6 L2 5 1

- c) Explain multithreading in Python with an example.

6 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
